

Problem

FAILURE MODE

Same scene. Same instruction. A camera-height or pitch change can still break a fixed-view policy.

Motivation

Viewpoint robustness should span camera height and pitch—without sacrificing standard-view performance.

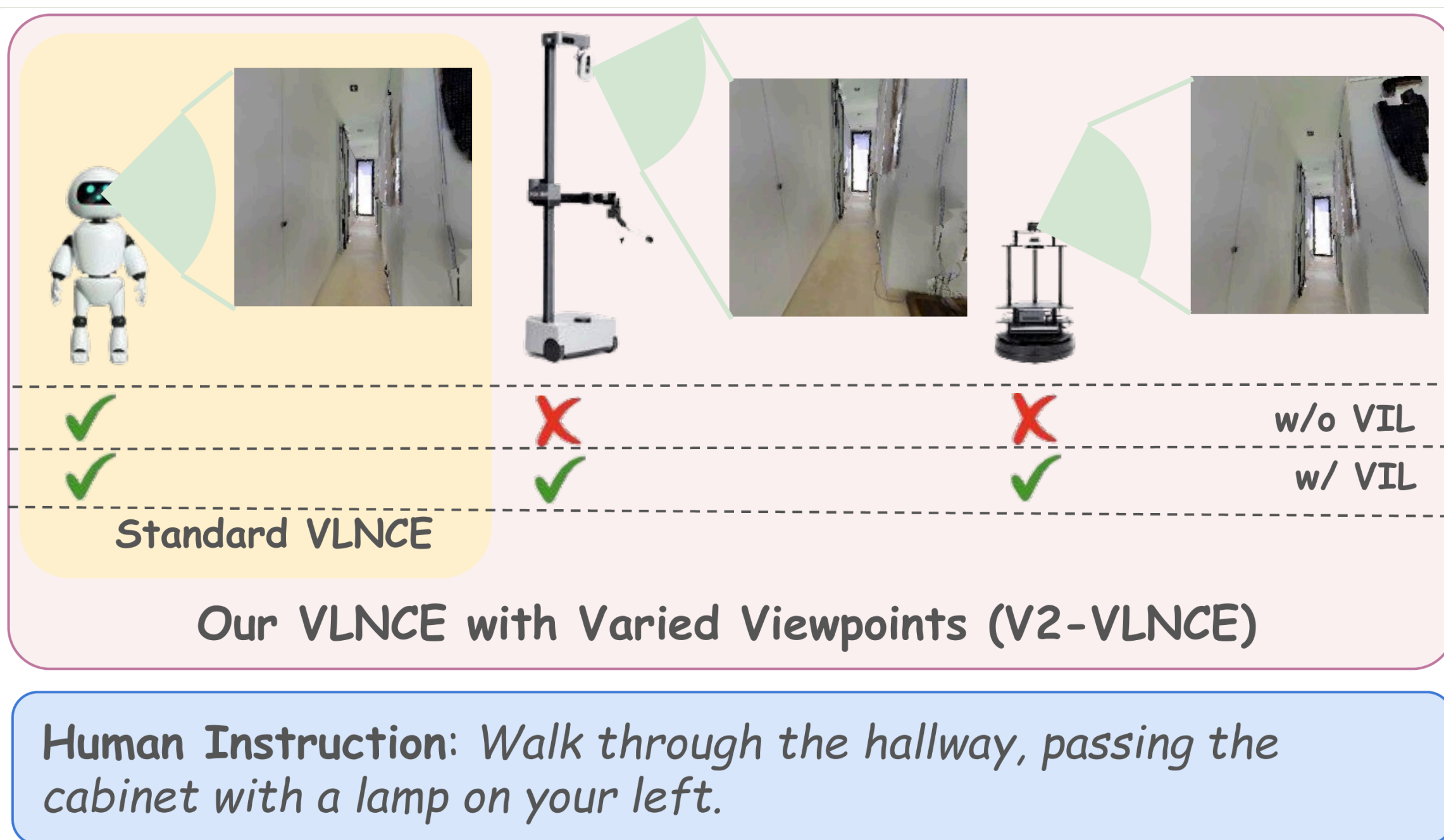
FIXED-VIEW TRAINING

Policies can fail as camera height and pitch shift.

VIEW-INVARIANT LEARNING

VIL adapts one pretrained navigator across the viewpoint distribution.

2-D viewpoint continuum. Camera height $\in [-0.5, 0.5]$ m and pitch $\in [-30^\circ, 30^\circ]$.



Human Instruction: Walk through the hallway, passing the cabinet with a lamp on your left.

Viewpoint shifts break the baseline; VIL restores successful navigation.

Method

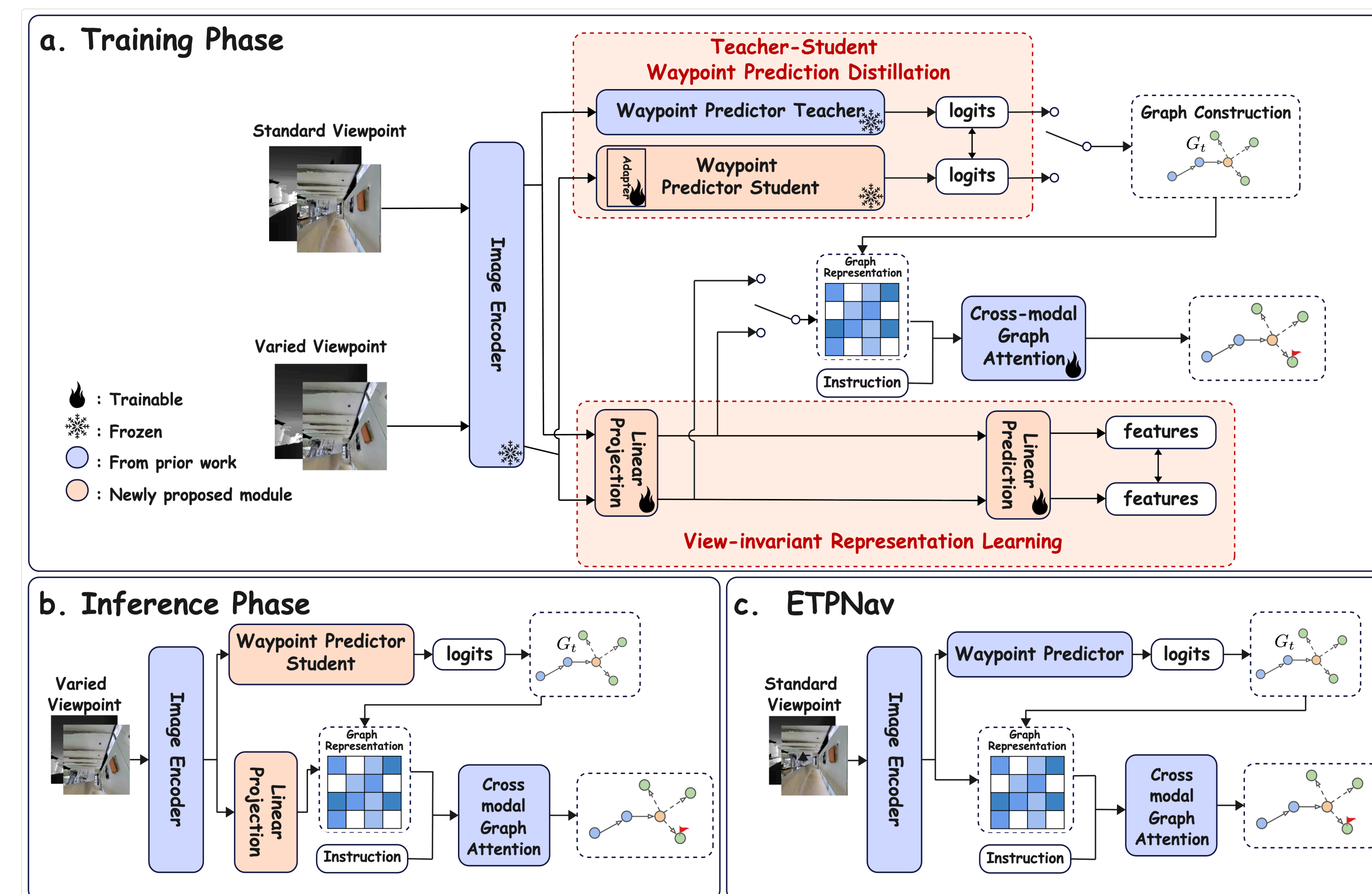
- 1 **Pair views** through one panoramic image encoder.
- 2 **Align features** with InfoNCE and hard negatives.
- 3 **Distill waypoints** from a frozen teacher into an adapter student.

$$\mathcal{L} = \mathcal{L}_{\text{nav}} + \lambda_1 \mathcal{L}_{\text{cl}} + \lambda_2 \mathcal{L}_{\text{wpd}}$$

Contrastive learning pulls together same-heading features from paired views while cross-scene and opposite-heading samples act as hard negatives.

Waypoint distillation transfers the frozen teacher's standard-view waypoint distribution to a lightweight student adapter through KL divergence.

No extra modules at inference. The teacher is frozen and discarded; only the compact waypoint student runs on the robot.



VIL jointly trains view-invariant features and waypoint distillation; only the student runs at inference.

Key Results

VARIED VIEW · VAL-UNSEEN

DATASET	METHOD	SR ↑	SPL ↑
R2R-CE	ETPNav	47	38
	+ VIL	55	45
RxR-CE	ETPNav	39	31
	+ VIL	50	41

R2R-CE SR +8–15 points
RxR-CE SR +11–15 points

STANDARD VIEW · VAL-UNSEEN

DATASET · METRIC	BASELINE	+ VIL
R2R-CE · SR	56.8	58.3
R2R-CE · SPL	48.9	49.7
RxR-CE · SR	53.8	55.6
RxR-CE · SPL	43.9	46.2

REMOVE WPD

Varied-view SPL

–8.3

Standard-view SPL

–4.1

+

REMOVE CL

Varied-view SPL

–2.8

Standard-view SPL

–1.0

CL and WPD both improve unseen-view generalization; WPD contributes the larger robustness gain.

Headline Numbers

+15 pts

Success Rate · varied viewpoints

R2R-CE / RxR-CE

Training Efficiency

14%

of baseline training time

48 hours vs. 11.5 days

Robot Deployment

PLATFORM

TurtleBot v2

SENSORS

RICOH THETA X 360°
RGB
Ouster OS0 Rev6
LiDAR

DEMO



TAKEAWAY

One navigation policy—robust across camera heights and pitches, with no standard-view trade-off.



CODE



HOMEPAGE